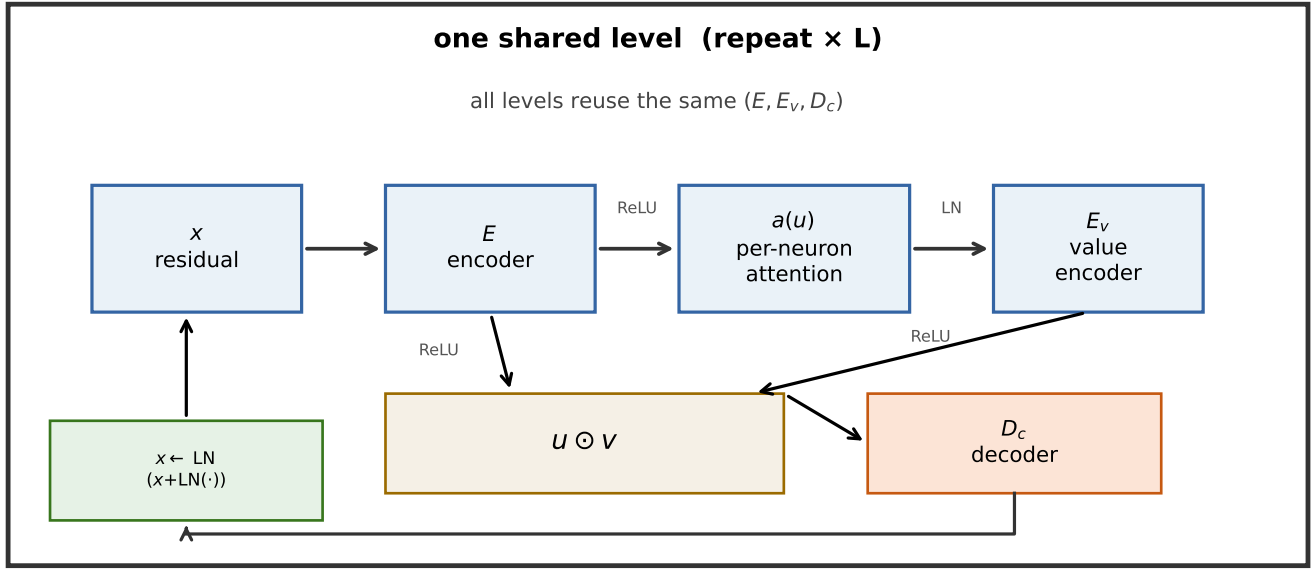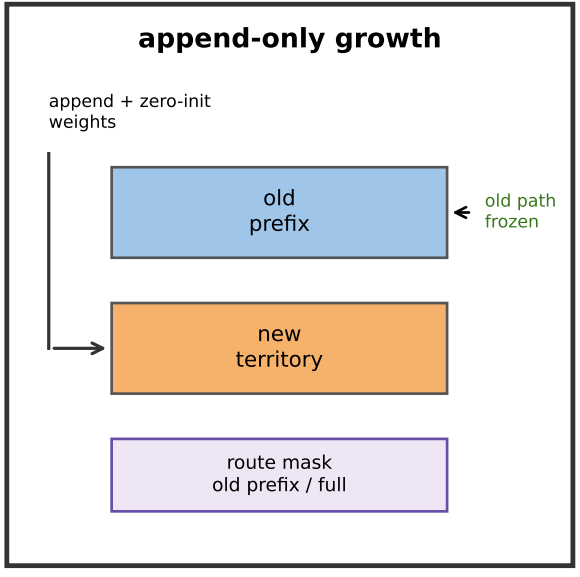BDH: shared depth-recurrent cell with append-only growth